



GRE PowerPrep Quantitative Reasoning – Study Review Guide

Section 3 of 5

Question 4 (Quantitative Comparison)

Problem Statement:

k is an integer for which:

$$\frac{1}{2^{1-k}} < \frac{1}{8}$$

- Quantity A: k
- Quantity B: -2

Options:

- (A) Quantity A is greater.
- (B) Quantity B is greater.
- (C) The two quantities are equal.
- (D) The relationship cannot be determined from the information given.

Correct Answer: (B) Quantity B is greater.

Step-by-Step Solution:

1. Note that $\frac{1}{8} = \frac{1}{2^3} = 2^{-3}$, and $\frac{1}{2^{1-k}} = 2^{-(1-k)} = 2^{k-1}$.
2. The inequality becomes: $2^{k-1} < 2^{-3}$
3. Since the exponential base $2 > 1$, the exponent inequality preserves the order: $k - 1 < -3$ implies $k < -2$
4. Since k is strictly less than -2 , Quantity A (k) is strictly less than Quantity B (-2).
5. Therefore, Quantity B is greater.

Data Set for Questions 6 to 8:

Table 1: Median Home Prices in the United States by Selected Region

Region	Median Home Price 1983	% Increase 1983 to 1986
Northeast	\$65,000	22.9%
West	\$97,000	11.3%



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South	\$92,000	11.0%
North Central	\$55,000	9.3%

Table 2: Median Home Prices and Related Data for Selected Northeastern Cities, 1986

(All figures are median amounts. Housing cost consists of mortgage payment, real estate taxes, utilities, and hazard insurance.)

City	Price of Home	Down Payment	Monthly Housing Cost	Annual Income
New York	\$129,700	\$42,250	\$1,141	\$49,692
Boston	\$126,000	\$30,000	\$1,116	\$55,956
Newark	\$116,000	\$29,450	\$1,139	\$54,660
Danbury	\$95,950	\$19,750	\$954	\$51,888
Nashua	\$82,000	\$10,650	\$891	\$43,200
Philadelphia	\$69,450	\$17,500	\$733	\$39,180
Trenton	\$66,000	\$16,500	\$703	\$37,434
Pittsburgh	\$54,150	\$11,500	\$673	\$40,680

Question 6 (Data Interpretation)**Problem Statement:**

In 1986, for which of the cities listed was the median home price minus the median down payment greatest?

Options:

- New York
- Boston
- Newark
- Danbury
- Pittsburgh

Correct Answer: Boston

Step-by-Step Solution:

Calculate $(\text{Median Home Price} - \text{Median Down Payment})$ for each given option:

- New York: $\$129,700 - \$42,250 = \$87,450$
- Boston: $\$126,000 - \$30,000 = \$96,000$
- Newark: $\$116,000 - \$29,450 = \$86,550$



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Question 7 (Data Interpretation)**Problem Statement:**

In 1986 the median home price in Danbury was approximately how many times the median home price in Trenton?

Options:

- 3
- 2
- $\frac{2}{3}$
- $\frac{1}{2}$
- $\frac{1}{3}$

Correct Answer: $\frac{1}{2}$

Step-by-Step Solution:

1. From Table 2, in 1986: Median home price in Danbury = \$95,950; Median home price in Trenton = \$66,000.
2. Ratio: $\frac{95,950}{66,000} \approx \frac{96,000}{66,000} = \frac{16}{11} \approx 1.455$.
3. Convert answer options to decimals: $\frac{1}{3} \approx 1.333$; $\frac{1}{2} = 1.500$; $\frac{2}{3} \approx 1.667$.
4. Difference comparisons: $|1.455 - 1.500| = 0.045$; $|1.455 - 1.333| = 0.122$.
5. \$1.455\$ is significantly closer to \$1.5\$ ($\frac{1}{2}$).

Question 8 (Data Interpretation)**Problem Statement:**

In 1986 in Newark, what was the approximate ratio of the median annual housing cost to the median homebuyer's annual income?

Options:

- $\frac{1}{7}$
- $\frac{1}{6}$
- $\frac{1}{4}$
- $\frac{1}{3}$
- $\frac{2}{5}$

Correct Answer: $\frac{1}{4}$

Step-by-Step Solution:

1. From Table 2, for Newark: Monthly housing cost = \$1,139 \implies Annual housing cost = $12 \times 1,139 = \$13,668$.
2. Homebuyer's annual income = \$54,660.

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Question 9 (Algebra / Exponents)

Problem Statement:

Which of the following is equal to $\frac{6^{14}}{(2^5)(3^7)}$?

Options:

- $(2^2)(3^2)$
- $(2^7)(3^7)$
- $(2^9)(3^2)$
- $(2^9)(3^7)$
- $(2^9)(3^9)$

Correct Answer: $(2^9)(3^7)$

Step-by-Step Solution:

1. Express the base 6 in terms of its prime factors: $6 = 2 \times 3$.
2. Thus, $6^{14} = (2 \times 3)^{14} = 2^{14} \times 3^{14}$.
3. Divide by the denominator: $\frac{2^{14} \times 3^{14}}{2^5 \times 3^7} = 2^{14-5} \times 3^{14-7} = 2^9 \times 3^7$.

Question 10 (Word Problem / Rates)

Problem Statement:

A certain train is traveling at a constant rate. If the train travels s miles in t hours, in how many hours will the train travel y miles?

Options:

- $\frac{sy}{t}$
- $\frac{st}{y}$
- $\frac{ty}{s}$
- $\frac{s}{ty}$
- $\frac{s+y}{t}$

Correct Answer: $\frac{ty}{s}$

Step-by-Step Solution:

1. Speed (Rate) = $\frac{\text{Distance}}{\text{Time}} = \frac{s}{t} \text{ mph}$.
2. Time needed to travel distance y = $\frac{\text{Distance}}{\text{Rate}} = \frac{y}{\frac{s}{t}} = y \times \frac{t}{s} = \frac{ty}{s} \text{ hours}$.

Question 11 (Probability)

Problem Statement:

A certain experiment has three possible outcomes. The outcomes are mutually exclusive and have probabilities p , $\frac{p}{2}$, and $\frac{p}{4}$, respectively. What is the value of p ?



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- $\frac{2}{7}$
- $\frac{3}{7}$
- $\frac{4}{7}$
- $\frac{5}{7}$

Correct Answer: $\frac{4}{7}$

Step-by-Step Solution:

1. Since the outcomes are mutually exclusive and account for all possible outcomes, their sum must equal 1: $p + \frac{p}{2} + \frac{p}{4} = 1$.
2. Find a common denominator (4): $\frac{4p + 2p + p}{4} = 1 \implies \frac{7p}{4} = 1$.
3. Multiply both sides by 4 and divide by 7: $7p = 4 \implies p = \frac{4}{7}$.

Question 12 (Geometry)

Problem Statement:

In the figure shown, two lines intersect forming an angle x° between them, with outer angles 30° and 40° at the upper corners, and a right angle (90°) at the bottom vertex. What is the value of x ?

Correct Answer: 160°

Step-by-Step Solution:

1. This figure is a non-convex (concave) quadrilateral.
2. By the exterior angle theorem, the interior non-reflex angle x equals the sum of the three opposite interior angles: $x = 30^\circ + 40^\circ + 90^\circ = 160^\circ$.
3. Alternatively, the four interior angles of the quadrilateral are 30° , 40° , 90° , and the reflex angle $(360^\circ - x)$.
4. $\text{Sum} = 30^\circ + 40^\circ + 90^\circ + (360^\circ - x) = 360^\circ \implies 160^\circ + 360^\circ - x = 360^\circ \implies x = 160^\circ$.

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Question 2 (Quantitative Comparison)

Problem Statement:

For all positive numbers p , the operation $\blacktriangledown$ is defined by:

$$p \blacktriangledown = p + \frac{1}{p}$$

- **Quantity A:** $(\frac{2}{7}) \blacktriangledown \blacktriangledown$
- **Quantity B:** 3.5

Options:

- (A) Quantity A is greater.

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Correct Answer: (A) Quantity A is greater.

Step-by-Step Solution:

1. Evaluate the inner operation for Quantity A: $(\frac{2}{7})^{\frac{1}{2}} = \frac{2}{7} + \frac{7}{2} = \frac{4 + 49}{14} = \frac{53}{14} \approx 3.786$.
2. Notice that $\frac{53}{14} > 3.5 = \frac{49}{14}$.
3. Apply $\frac{1}{2}$ a second time: $(\frac{53}{14})^{\frac{1}{2}} = \frac{53}{14} + \frac{14}{53} \approx 3.786 + 0.264 = 4.05$.
4. Since $4.05 > 3.5$, Quantity A is strictly greater than Quantity B.

Question 4 (Quantitative Comparison)

Problem Statement:

In triangle IGK :

- H is the midpoint of IG
- J is the midpoint of IK
- $IG = GK = 10$
- **Quantity A:** IJ
- **Quantity B:** 5

Options:

- (A) Quantity A is greater.
- (B) Quantity B is greater.
- (C) The two quantities are equal.
- (D) The relationship cannot be determined from the information given.

Correct Answer: (D) The relationship cannot be determined from the information given.

Step-by-Step Solution:

1. Since J is the midpoint of IK , $IJ = \frac{1}{2} IK$.
2. By the Triangle Inequality theorem in $\triangle IGK$: $|IG - GK| < IK < IG + GK \implies |10 - 10| < IK < 10 + 10 \implies 0 < IK < 20$.
3. Dividing by 2 gives: $0 < IJ < 10$.
4. If $IK = 10$, then $IJ = 5$ ($A = B$). If $IK = 14$, $IJ = 7$ ($A > B$). If $IK = 6$, $IJ = 3$ ($B > A$).
5. Therefore, the relationship cannot be determined.

Question 5 (Quantitative Comparison)

Problem Statement:

The figure shown is a regular octagon. A diagonal of an octagon is any line segment connecting two nonadjacent vertices.

- **Quantity A:** The number of diagonals of the octagon that are parallel to at least one side



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- (A) Quantity A is greater.
- (B) Quantity B is greater.
- (C) The two quantities are equal.
- (D) The relationship cannot be determined from the information given.

Correct Answer: (B) Quantity B is greater.

Step-by-Step Solution:

1. Total number of diagonals in an n -sided polygon is $\frac{n(n-3)}{2}$. For $n = 8$: $\frac{8 \times 5}{2} = 20$.
2. In a regular octagon, there are 4 pairs of parallel sides. For each pair of opposite sides, exactly 2 diagonals are parallel to those sides.
3. Quantity A = $4 \times 2 = 8$.
4. Quantity B = $20 - 8 = 12$.
5. Since $12 > 8$, Quantity B is greater.

Question 7 (Geometry)

Problem Statement:

The perimeter of square SSS is 40. Square TTT is inscribed in square SSS . What is the least possible area of square TTT ?

Options:

- 45
- 48
- 49
- 50
- 52

Correct Answer: 50

Step-by-Step Solution:

1. Perimeter of $SSS = 40 \implies$ side length = 10.
2. Let a and $10 - a$ be segments of the side of SSS divided by a vertex of TTT .
3. $\text{Area}(T) = a^2 + (10 - a)^2 = 2a^2 - 20a + 100$.
4. The minimum value occurs when $a = 5$ (midpoint): $\text{Area}_{\min} = 2(5)^2 - 20(5) + 100 = 50 - 100 + 100 = 50$.

Question 8 (Fractions / Word Problem)

Problem Statement:

Kelly took three days to travel from City C to City D. On the first day, Kelly traveled $\frac{2}{5}$ of the distance. On the second day, she traveled $\frac{2}{3}$ of the remaining distance. Which is the fraction traveled on the third day?



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- $1 - \frac{2}{5} - \frac{2}{3} \left(\frac{2}{5} \right)$
- $1 - \frac{2}{5} - \frac{2}{5} \left(1 - \frac{2}{3} \right)$
- $1 - \frac{2}{5} - \frac{2}{3} \left(1 - \frac{2}{5} \right)$
- $1 - \frac{2}{5} - \frac{2}{3} \left(1 - \frac{2}{5} - \frac{2}{3} \right)$

Correct Answer: $1 - \frac{2}{5} - \frac{2}{3} \left(1 - \frac{2}{5} \right)$

Step-by-Step Solution:

1. Day 1: $\frac{2}{5}$. Remaining distance: $1 - \frac{2}{5}$.
2. Day 2: $\frac{2}{3}$ of $\left(1 - \frac{2}{5} \right) = \frac{2}{3} \left(1 - \frac{2}{5} \right)$.
3. Day 3: $1 - (\text{Day 1}) - (\text{Day 2}) = 1 - \frac{2}{5} - \frac{2}{3} \left(1 - \frac{2}{5} \right)$.

Question 9 (Number Properties)

Problem Statement:

If x and y are integers and $x = 50y + 69$, which of the following must be odd?

Options:

- xy
- $x + y$
- $x + 2y$
- $3x - 1$
- $3x + 1$

Correct Answer: $x + 2y$

Step-by-Step Solution:

1. $x = 50y + 69$. Since $50y$ is even and 69 is odd, x is always odd.
2. Evaluate choices: $x + 2y = \text{odd} + \text{even} = \text{odd}$. This is always odd.

Question 10 (Weighted Averages / Percentages)

Problem Statement:

A team won 60% of its games in the first half. In the second half, it played 20 games and won 3. If the team won 50% of the games total last year, what was the total number of games played?

Options:

- 60
- 70
- 80
- 90
- 100

Correct Answer: 90



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2. Second half: 20 games, 3 wins.
3. Total games = $\$g + 20\$$. Total wins = $\$0.60g + 3\$$.
4. $\$0.60g + 3 = 0.50(g + 20) \implies 0.60g + 3 = 0.50g + 10 \implies 0.10g = 7 \implies g = 70\$$.
5. Total games = $\$70 + 20 = 90\$$.

Question 11 (Rounding / Integer Constraints)**Problem Statement:**

Of 1,150 people, $\$x\$$ percent chose brand D. If $\$x\$$ rounded to the nearest integer is 3, which could be the number of people who chose D?

Options:

- 20, 25, 30, 35, 40, 45, 50

Correct Answer: 30, 35, 40

Step-by-Step Solution:

1. $\$2.5 \leq x < 3.5\$$.
2. $\$x = \frac{N}{1150} \times 100 = \frac{N}{11.5} \$$.
3. $\$2.5 \times 11.5 \leq N < 3.5 \times 11.5 \implies 28.75 \leq N < 40.25\$$.
4. Valid integers: 29 through 40. Options matching: 30, 35, 40.

Question 12 (Sequences / Telescoping Series)**Problem Statement:**

$\$b_k = \frac{1}{k} - \frac{1}{k+1} \$$. What is the sum of the first 100 terms?

Correct Answer: $\frac{100}{101} \$$

Step-by-Step Solution:

1. Sum = $\$(1 - \frac{1}{2}) + (\frac{1}{2} - \frac{1}{3}) + \dots + (\frac{1}{100} - \frac{1}{101})\$$.
2. Intermediate terms cancel: $\$1 - \frac{1}{101} = \frac{100}{101} \$$.

Question 13 (Absolute Value Inequalities)**Problem Statement:**

If $\$|x + 1| \leq 5\$$ and $\$|y - 1| \leq 5\$$, what is the least possible value of $\$xy\$$?

Correct Answer: $\$-36\$$

Step-by-Step Solution:

1. $\$ -6 \leq x \leq 4 \$$ and $\$ -4 \leq y \leq 6 \$$.
2. Least product $\$xy\$$: $\$(-6) \times (6) = -36\$$.



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Problem Statement:

City B revenue: 1990 to 1995 was -15% change; 1995 to 2000 was $+15\%$ change. If 1990 revenue was $\$800,000$, what was it in 2000?

Correct Answer: $\$782,000$

Step-by-Step Solution:

1. Revenue (1995) = $\$800,000 \times 0.85 = 680,000$.
2. Revenue (2000) = $\$680,000 \times 1.15 = 782,000$.

Question 15 (Inequalities / Positive Numbers)**Problem Statement:**

If $x, y, z > 0$ and $3x < 2y < 4z$, which could be true?

Options:

- $x = y$, $y = z$, $y > z$, $x > z$

Correct Answer: $y = z$, $y > z$, $x > z$

Step-by-Step Solution:

1. $x = y$: $3x < 2x \implies 3 < 2$ (False).
2. $y = z$: Let $y=1, z=1 \implies 2 < 4$. If $x=0.5, 1.5 < 2 < 4$ (True).
3. $y > z$: Let $z=1, y=1.5 \implies 3 < 4$. If $x=0.5, 1.5 < 3 < 4$ (True).
4. $x > z$: Let $z=1, x=1.1, y=1.8 \implies 3.3 < 3.6 < 4$ (True).